

2025-26 Archimedean Satellite Invitational Meteorology Exam

This test contains 79 questions and a combination of many easy questions to some more difficult analysis questions. Please make sure you can access the image sheet at the beginning of the test as most questions refer to it. Because there is an image sheet, time outside the browser will not be counted. Please keep your answers concise for ease of grading, but answer to the best of your ability. Email me at om.guin@gmail.com if you have any questions. Good luck!

1. [1] (SATA) In which of the following locations exist temperature inversions?
    a. Tropopause
    b. Stratopause
    c. Thermopause
    d. Exopause
2. [1] (MCQ) Which of the following layers of the atmosphere has the highest temperature?
    a. Troposphere
    b. Stratosphere
    c. Mesosphere
    d. Thermosphere
    e. Exosphere
3. [1] (MCQ) Which of the following boundaries define the “edge” of the atmosphere?
    a. Exopause
    b. Karman Line
    c. Bergeron Line

    d. Balmer Line
4. [1] (MCQ) According to Earth's energy budget, how much energy is absorbed by land and oceans?
    a. 16%
    b. 89%
    c. 51%
    d. 23%
5. [1] (MCQ) Which of the following does NOT determine how much insolation is received at a surface?
    a. Latitude
    b. Length of the day
    c. Transparency of the atmosphere
    d. Longitude
6. [1] (MCQ) Wet soil changes temperature at a slower rate than dry soil. This means wet soil has a ___ average heat capacity than dry soil.
    a. Lower
    b. Higher
    c. Same
    d. Unrelated
7. [2] (FRQ) On a very cold day, a puddle of water gradually cools to ice just before you stick a thermometer in it. You observe that the puddle was at 0 degrees Celsius before and after it froze and also observe an increase in the temperature of the air above the puddle. Explain this increase in temperature and what type of heat caused it.
8. [2] (MCQ) Which of the following is NOT indicative of a 100% relative humidity?
    a. Evaporation no longer occurs
    b. The dry bulb temperature is equal to the wet bulb temperature
    c. Fog begins to form
    d. Virga forms
9. [1] (FRQ) Explain the effect of increasing the axial tilt of Earth on the intensity of the seasons.
10. [2] (FITB) Earth is ____ (closer/farther) to the sun in the winter and ____ (closer/farther) from the sun in the summer.
11. [1] (FITB) Cyclogenesis is the ____ (formation/dissipation) of a cyclone while cyclolysis is the ____ (formation/dissipation) of a cyclone.
12. [3] (FITB) Cyclones rotate ____ (CW/CCW) in the NH, and anticyclones rotate ___ (CW/CCW) in the SH. This is due to the ____ Effect.
13. [2] (FITB) Compared to valley winds, mountain winds are ____ (anabatic/katabatic), ____ (denser/less dense), and ____ (cause/don't cause) fog.

14. [2] (FITB) Sea breezes occur during the ____ (day/night) while land breezes occur during the ____ (day/night).
15. [1] (MCQ) The warming of a chinook wind as it moves downslope is an ____ process.
    a. Isothermal
    b. Isobaric
    c. Adiabatic
    d. Expansionary
16. [1] (MCQ) Which of the following cloud types would you expect to reflect the most solar radiation away from Earth?
    a. Altostratus
    b. Cirrocumulus
    c. Cumulonimbus
    d. Stratus
17. [1] (MCQ) Which of the following cloud types would you expect to "trap" the most solar radiation?
    a. Altocumulus
    b. Cirrus
    c. Cumulonimbus
    d. Cumulus
18. [4] (FRQ) The air temperature is generally above the dew point. As expected, as the air temperature decreases towards the dew point, the relative humidity increases. However, the amount of water vapor in the air can stay the same during this process. Why is this possible? In other words, why would the relative humidity increase as temperature decreases even though the water vapor content remains the same?
19. [3] (FRQ) We know that temperature decreases as altitude increases in the troposphere. Therefore, the relative humidity of a rising parcel of air in the troposphere will keep increasing in relative humidity until its water vapor condenses. What is the name of the altitude at which this condensation first occurs?
20. [2] (FRQ) The previous question assumes that the air parcel is rising. Identify and explain two reasons an air parcel might lift.

Look at the image sheet. Identify cloud types depicted in images 1A-1J. Please use the following acronyms for ease of grading.
Cirrus: Ci
Cirrocumulus: Cc
Cirrostratus: Cs
Altocumulus: Ac
Altostratus: As

Nimbostratus: Ns
Cumulus: Cu
Stratocumulus: Sc
Stratus: St
Cumulonimbus: Cb
Lenticular: Lc
Pyrocumulonimbus: PCb

21. [0.5] (SA) Image 1A
22. [0.5] (SA) Image 1B
23. [0.5] (SA) Image 1C
24. [0.5] (SA) Image 1D
25. [0.5] (SA) Image 1E
26. [0.5] (SA) Image 1F
27. [0.5] (SA) Image 1G
28. [0.5] (SA) Image 1H
29. [0.5] (SA) Image 1I
30. [0.5] (SA) Image 1J

Look at the image sheet. Identify the atmospheric optical effects shown in Images 2A-2H.

31. [0.5] (SA) Image 2A
32. [0.5] (SA) Image 2B
33. [0.5] (SA) Image 2C
34. [0.5] (SA) Image 2D
35. [0.5] (SA) Image 2E
36. [0.5] (SA) Image 2F
37. [0.5] (SA) Image 2G
38. [0.5] (SA) Image 2H

Look at the image sheet. Identify the weather instruments in images 3A-3H

39. [0.5] (SA) Image 3A
40. [0.5] (SA) Image 3B
41. [0.5] (SA) Image 3C
42. [0.5] (SA) Image 3D
43. [0.5] (SA) Image 3E
44. [0.5] (SA) Image 3F
45. [0.5] (SA) Image 3G
46. [0.5] (SA) Image 3H

Refer to the surface weather map shown in image 4 for questions 47 - 52

47. [2] (FRQ) Which type of front is forming near the low pressure cell north of Wisconsin?
48. [2] (FRQ) What is the approximate pressure, in mb, around North Dakota?
49. [4] (FRQ) Calculate the average pressure gradient (SI units) from South Dakota to North Wisconsin. Assume the distance is roughly 700 miles. Show your work.
50. [2] (FRQ) It is evident that the actual pressure gradient from the previous question is not uniform. In other words, you calculated the average pressure gradient. Where (closer to SD or WI) would you expect a steeper gradient? Explain.
51. [2] (FRQ) What type of clouds, if any, would you expect near North Utah to South Wyoming? Explain.
52. [3] (FRQ) Which region is experiencing the strongest cold air advection? Explain.

Refer to the surface weather map shown in image 5 for questions 53 - 57.

53. [2] (FRQ) There is an occluded front in the Northeastern US. Which state is the triple point located in?
54. [2] (FRQ) Is this occlusion intensifying or weakening? How can you tell?
55. [3] (FRQ) Which air masses are interacting with this occluded front? Where is each from?
56. [1] (FRQ) There is a blue dashed line around the Great Plains region. What does this line signify?
57. [1] (FRQ) What are the orange dashed lines called?

Refer to the 500 mb upper air chart shown in image 6 for questions 58 - 62

58. [1] Is the flow over most of the US more zonal or meridional?
59. [1] Which region has the strongest 500-mb winds?
60. [2] Would you expect rising air more under a ridge or under a trough?
61. [6] Identify the location of any troughs or ridges. For each one, identify whether it is positively tilted, neutral, or negatively tilted. Clearly articulate the location by stating which states or parts of the image are covered.
62. [3] Over which region would you expect the greatest large-scale upward motion from positive vorticity advection? Explain.

63. [2] (FRQ) Refer to image 7. Each panel (A–D) shows a vertical atmospheric temperature profile (red line) versus altitude, with the black vertical line marking 0°C. For each profile, determine the precipitation type expected at the ground.

Refer to the station model shown in image 8 for questions 64 - 72

64. [0.5] (SA) What is the temperature, in celsius, in this region?

65. [0.5] (SA) What is the dew point, in Fahrenheit, in this region?
66. [0.5] (SA) What percentage of the sky is covered?
67. [1] (SA) What is the wind speed in mph?
68. [0.5] (SA) What is the barometric pressure in mb?
69. [1] (SA) What was the pressure in mb 3 hours ago?
70. [1] (SA) What type of cloud is the low cloud? (use abbreviation)
71. [1] (SA) What type of cloud is the high cloud? (use abbreviation)
72. [1] (SA) What intensity level is the rain?

Refer to the following METAR code for questions 73 - 79

METAR KJFK 301651Z 22018G28KT 10SM -RA BKN020 OVC070 12/09 A2992 RMK AO2 SLP134 T01220094

73. [1] (FRQ) What day of the month and at what time was this taken?
74. [1] (FRQ) Where is the wind blowing from and at what speed? Include units
75. [1] (FRQ) What is the visibility? Include units
76. [2] (FRQ) What are the altitudes of the broken and overcast cloud layers, in feet above ground level?
77. [2] (FRQ) What is the temperature and dew point in celsius?
78. [2] (FRQ) What is the pressure in inHg?
79. [2] (FRQ) What is the sea level pressure in hPa?